

# DBMS Normalization Notes

## What is Normalization?

Normalization is the process of organizing data in a database to reduce redundancy, avoid anomalies, and improve data integrity.

## Why Normalization is Needed?

Normalization helps prevent insertion anomalies, update anomalies, and deletion anomalies.

## First Normal Form (1NF)

A table is in 1NF if each column contains atomic values and there are no repeating groups.

## Second Normal Form (2NF)

A table is in 2NF if it is already in 1NF and has no partial dependency.

## Third Normal Form (3NF)

A table is in 3NF if it is already in 2NF and has no transitive dependency.

## Boyce-Codd Normal Form (BCNF)

BCNF states that for every functional dependency, the determinant must be a super key.

## Fourth Normal Form (4NF)

A table is in 4NF if it is in BCNF and has no multi-valued dependency.

## Fifth Normal Form (5NF)

A table is in 5NF if there are no join dependencies.

## Functional Dependency

Functional Dependency means one attribute uniquely determines another attribute. Example: StudentID → StudentName

## Advantages of Normalization

Reduces redundancy, improves consistency, saves storage, and improves maintenance.

## Disadvantages of Normalization

Creates more tables and increases the complexity of joins.

## Quick Comparison Table

| Normal Form | Removes                  |
|-------------|--------------------------|
| 1NF         | Repeating Groups         |
| 2NF         | Partial Dependency       |
| 3NF         | Transitive Dependency    |
| BCNF        | Non-super key dependency |
| 4NF         | Multi-valued dependency  |
| 5NF         | Join dependency          |

## Important Exam Questions

- What is normalization?
- Explain anomalies in DBMS.
- Difference between 2NF and 3NF.
- What is BCNF?
- Explain functional dependency.
- What is transitive dependency?
- Difference between candidate key and primary key.